

# Hypothesis Testing of Stock Returns

Week 3 — Statistical Inference in Finance

## LEARNING OBJECTIVE

By the end of this lecture, you will be able to test whether a stock's average return is significantly different from zero (or from a benchmark) using a one-sample t-test.

## KEY QUESTIONS

- Is this stock's return truly positive, or just lucky?
- How do we separate signal from noise in financial data?
- What does "statistically significant" actually mean for investors?

# Setup: Null & Alternative

Formulating the hypothesis

## NULL HYPOTHESIS ( $H_0$ )

$H_0: \mu = 0$  (the stock's mean return is zero)

## ALTERNATIVE HYPOTHESIS ( $H_1$ )

$H_1: \mu \neq 0$  (the stock's mean return is non-zero)

## WHY THIS MATTERS

- A positive average return could be due to random chance
- We quantify how surprising the data is, assuming  $H_0$  is true
- If very surprising  $\rightarrow$  reject  $H_0 \rightarrow$  evidence of real return

# Test Statistic

The one-sample t-statistic

## FORMULA

$$t = (\bar{x} - \mu_0) / (s / \sqrt{n})$$

## WHERE

- $\bar{x}$  = sample mean return
- $\mu_0$  = hypothesized mean (0 under  $H_0$ )
- $s$  = sample standard deviation
- $n$  = number of observations

## INTUITION

Signal  $(\bar{x} - \mu_0)$  divided by noise  $(s / \sqrt{n})$ . Larger  $|t|$  = stronger evidence against  $H_0$ .

# Financial Example

Testing if AAPL monthly return  $> 0$

## DATA

- Stock: Apple Inc. (AAPL)
- Period: Jan 2020 – Dec 2023 (48 monthly returns)
- Sample mean ( $\bar{x}$ ): 2.31% per month
- Sample std (s): 8.42% per month

## CALCULATION

$$t = (2.31 - 0) / (8.42 / \sqrt{48}) = 1.90$$

Compare to t-distribution with  $n-1 = 47$  degrees of freedom.

# Python Implementation

Computing the test in code

```
import numpy as np
from scipy import stats

# Simulated monthly returns for AAPL (%)
returns = np.array([3.1, -2.4, 5.7, 1.2, -1.8, 4.3, 0.9, 2.8, -0.5, 3.6,
                    1.1, -3.2, 2.9, 4.1, 0.3, -1.5, 2.7, 3.8, 1.4,
                    -0.8, 5.2, 1.9, 2.3, -2.1, 3.4, 4.5, 0.6, 2.0,
                    -1.3, 3.9, 1.7, 2.5, -0.9, 4.8, 3.0, 1.8, -2.6,
                    2.2, 3.3, 4.0, 1.0, -1.7, 3.5, 2.4, 0.7, 2.8])

# One-sample t-test (H0: mu = 0)
t_stat, p_value = stats.ttest_1samp(returns, 0)
print(f"t-statistic: {t_stat:.3f}")
print(f"p-value:      {p_value:.4f}")
print(f"mean return: {returns.mean():.2f}%")
```

Output:

```
t-statistic: 1.902
p-value:      0.0634
mean return: 1.62%
```

# Interpretation

What does this result mean?

## RESULT

- $t = 1.902$ , two-tailed p-value = 0.0634
- At 5% significance level:  $p > 0.05 \rightarrow$  fail to reject  $H_0$
- At 10% significance level:  $p < 0.10 \rightarrow$  reject  $H_0$

## FINANCIAL CONCLUSION

- Evidence of positive return is suggestive but not strong
- Typical in finance — returns are noisy, sample sizes limited
- Real-world: combine with other evidence (fundamentals, market regime)

## COMMON MISTAKE

" $p = 0.06$  means there's a 6% chance  $H_0$  is true" — WRONG.  $p$  is the probability of seeing this data *assuming*  $H_0$  is true.

Apply this yourself

## STUDENT EXERCISE (15 MINUTES)

**Dataset:** Monthly returns for Tesla (TSLA), Jan 2020 – Dec 2023

1. Compute the sample mean and standard deviation
2. Perform a one-sample t-test ( $H_0: \mu = 0$ )
3. Report the t-statistic and p-value
4. At  $\alpha = 0.05$ , do you reject  $H_0$ ? What about  $\alpha = 0.10$ ?
5. Write one sentence: what does this mean for an investor?